



DEPARTMENT OF COMPUTER SCIENCE & IT

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A project submitted in partial fulfillment for the requirement of the award of Bachelor of (insert degree name) Degree of Kabarak University.

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[Benedict Kipkorir] [CS/MK/0700/09/23] [March 2026]

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DEDICATION

This project is lovingly dedicated to my parents and family, whose endless sacrifices, prayers, encouragement, and unwavering belief in my abilities have carried me through every step of this academic journey. It is also dedicated to every Kenyan consumer navigating the world of online shopping in search of better value and smarter choices—may tools like this one bring a little more ease and fairness to your daily decisions.

ABSTRACT

The rapid expansion of e-commerce has increased online shopping on platforms such as Jumia, Kilimall, Amazon, etc., yet consumers struggle with manual price comparison, tracking trends, and identifying genuine deals due to dynamic pricing and platform fragmentation. This project develops “.....”, an automated web scraping tool that extracts real-time product data (prices, ratings, availability) from selected e-commerce sites, performs side-by-side comparisons, visualizes price history, and delivers timely alerts for price drops.

The approach includes market analysis to identify user challenges, Python-based scraping with Requests, BeautifulSoup, and Selenium (with ethical practices like rate limiting), data storage in SQLite, processing via Pandas, and a Streamlit dashboard for interactive visualization. Automation uses APScheduler for scheduled tasks and notifications.

Results show the tool effectively reduces search time and supports informed decisions through accurate comparisons and trend tracking. The project concludes that such automated solutions can empower consumers in the digital marketplace, though ongoing maintenance is needed for evolving website structures.

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CHAPTER ONE

INTRODUCTION

1.1 Introduction

This chapter lays the foundation for the study by providing the background and context of the project, identifying the core problem it seeks to address, and articulating its purpose and value. It begins with an overview of the rapid expansion of e-commerce, driven by increasing internet penetration, mobile money adoption (such as M-Pesa), and platforms like Jumia and Kilimall. Recent data indicate that Kenya's B2C e-commerce market is projected to reach approximately US\$2.61 billion by 2025, with continued growth expected through 2029 at a CAGR of around 10%. Despite this progress, consumers face persistent challenges in navigating price variations across platforms, manually comparing deals, and identifying genuine discounts amid dynamic pricing and promotional fluctuations.

The chapter then defines the specific problem arising from these inefficiencies in current online shopping practices. It outlines the project's main and specific objectives, which guide the development of an automated tool to streamline price monitoring and decision-making. The proposed system modules are introduced briefly, highlighting their roles in solving the identified issues. Finally, the significance of the project, its scope, and limitations are discussed to establish clear boundaries. Overall, Chapter One justifies the need for this work and sets the direction for subsequent chapters by presenting the key research questions and objectives.

1.2 Background of the Study

The e-commerce sector has experienced significant growth in recent years, fueled by widespread mobile internet access, a young and tech-savvy population, and innovative digital payment systems. As of recent estimates, the market is valued at around US\$2.31 billion in 2024, with projections indicating annual growth of 12.9% to reach US\$2.61 billion by 2025 and further expansion toward US\$3.83 billion by 2029. Major platforms such as Jumia and Kilimall dominate the landscape, offering a wide range of products including electronics, fashion, and home appliances. These platforms serve millions of users, with a notable portion of Kenyan internet users (approximately 37.6%) engaging in online purchases and about 15.7% utilizing price comparison services.

For consumers, particularly in urban centers like Nairobi, these platforms provide convenience and variety. However, operations remain fragmented: sellers on different sites set independent prices, run frequent promotions, and adjust dynamically based on demand, stock, or competition. This leads to noticeable price differences for identical or similar products (e.g., electronics often listed cheaper on Kilimall compared to Jumia in user observations). Shoppers currently rely on manual browsing across multiple tabs or apps, which is time-consuming and prone to missing better deals. There is no centralized, automated mechanism widely accessible to ordinary

consumers for real-time cross-platform comparison, historical tracking, or proactive alerts on price drops.

This project emerges in response to these consumer-level inefficiencies within Kenya's evolving e-commerce ecosystem, aiming to empower users with data-driven tools to make informed purchasing decisions.

1.3 Problem Statement

Despite the rapid growth of e-commerce, consumers encounter several challenges stemming from current online shopping operations on platforms like Amazon, Jumia and others. These include:

- **Time-consuming manual price comparison** — Shoppers must open multiple websites or apps, search for the same product repeatedly, and compare prices, descriptions, and ratings manually, leading to inefficiency and decision fatigue, especially for price-sensitive buyers.
- **Inability to track price history and detect genuine deals** — Dynamic pricing and flash sales make it difficult to determine if a current price represents a real discount or a temporary fluctuation, resulting in potential overpayment.
- **Lack of timely notifications for price drops** — Without automated monitoring, consumers miss opportunities for savings when prices fall below a preferred threshold, particularly in a market where price sensitivity is high among different socio-economic groups.
- **Fragmented information across platforms** — Variations in pricing, availability, and promotions between sites (e.g., slight differences noted between Jumia and Kilimall for the same items) are not aggregated, forcing users to rely on incomplete or outdated mental comparisons.

These problems originate directly from the decentralized and competitive nature of e-commerce operations, where platforms operate independently without integrated consumer tools for cross-site intelligence. This project addresses these gaps by developing an automated web scraper to enable efficient, real-time comparison and alerts, contributing to more informed and cost-effective online shopping.

1.4 Objectives

1.4.1 Main Objective

The main objective of this project is to develop an automated web scraping tool for real-time e-commerce price comparison and deal alerts, enabling consumers to efficiently identify the best deals across multiple platforms within a single interface.

1.4.2 Specific Objectives

- (i) To investigate the current challenges and limitations faced by online shoppers in comparing prices and tracking deals across major e-commerce platforms.
- (ii) To design and develop a Python-based web scraper capable of extracting real-time product data (prices, ratings, availability) from selected e-commerce websites while adhering to ethical scraping practices.
- (iii) To implement data storage, processing, and visualization features that enable price history tracking, side-by-side comparisons, and trend analysis for informed decision-making.
- (iv) To integrate automation and notification mechanisms (e.g., scheduled scraping and email/SMS alerts) to notify users of significant price drops or better deals in a timely manner.

1.5 Research Questions

- (i) What are the key challenges and limitations encountered by consumers when manually comparing prices and tracking deals across major e-commerce platforms?
- (ii) How can a Python-based web scraper be designed and implemented to reliably extract real-time product information from selected e-commerce sites while maintaining ethical and technical standards?
- (iii) What data processing and visualization techniques can be applied to store, analyze, and present price history and comparisons effectively for user decision support?
- (iv) How can automation and notification systems be integrated to provide timely alerts on price changes and optimal deals to enhance the user experience?

1.6 Significance of the Study

This project holds significant value in the context of the booming e-commerce sector, where price sensitivity remains a dominant factor in consumer behavior. By automating price comparison and deal detection, the tool addresses a timely need for efficiency in a market characterized by platform fragmentation and dynamic pricing. Its realization offers practical advantages: consumers can save time and money, reduce decision fatigue, and make more confident purchases, particularly amid economic pressures where finding the best value is critical.

The project is challenging and interesting from a technical standpoint, involving web scraping ethics, data handling, automation, and user interface design—skills highly relevant to modern software development. It contributes to digital empowerment by providing an accessible, educational tool (for personal/non-commercial use) that promotes transparency in online retail. Academically, it adds to local literature on consumer-facing e-commerce innovations in developing markets. Ultimately, successful implementation could inspire similar tools or

extensions (e.g., mobile apps), benefiting individual shoppers and indirectly supporting fairer competition among platforms.

1.7 Scope and Limitation of the Study

Scope: The project focuses on developing a prototype web scraper targeting 2–3 major e-commerce platforms (primarily Jumia and Kilimall, with potential inclusion of others like Amazon if feasible). It covers real-time data extraction (price, basic details), storage in a local database, basic comparison/visualization via a web dashboard, and automated alerts. The timeframe for development and testing is within the academic year (2025–2026). The system targets individual consumers for personal use.

Limitations: The tool relies on public webpage structures, which may break if sites update their HTML/CSS (requiring maintenance). It does not use official APIs (due to access restrictions) and adheres strictly to ethical guidelines (rate limiting, robots.txt respect), potentially limiting scraping frequency or depth. Legal risks associated with scraping (even public data) exist. Features like multi-user accounts, advanced ML predictions, or mobile app integration are beyond scope. Performance depends on internet stability and may vary in rural areas with poor connectivity.

1.8 Proposed Modules

Module 1: Web Scraping Module This core module handles data extraction from target e-commerce websites. It includes:

- Product search and URL generation based on user input.
- Parsing of HTML to retrieve price, product name, rating, availability, and link.
- Error handling for failed requests, dynamic content (using Selenium if needed), and anti-scraping measures (delays, user-agent rotation). Technology: Python with Requests/BeautifulSoup for static pages and Selenium for JavaScript-rendered content.

Module 2: Data Storage and Processing Module This module manages persistent storage and analysis of scraped data. It includes:

- Insertion of timestamped records into a database.
- Cleaning and normalization (e.g., currency handling, duplicate removal).
- Computation of comparisons (min price, savings percentage) and history trends. Technology: SQLite (or MongoDB) for storage, Pandas for processing.

Module 3: User Interface and Visualization Module This module provides an interactive frontend for users. It includes:

- Search input form and display of real-time comparisons in tables/charts.
- History graphs (price trends over time).
- Dashboard for managing watchlists. Technology: Streamlit for rapid web-based UI and Plotly for interactive visualizations.

Module 4: Automation and Notification Module This module enables scheduled operations and user alerts. It includes:

- Background scraping at set intervals.
- Threshold-based checks for price drops.
- Sending email or SMS notifications. Technology: APScheduler for timing, smtplib/Twilio for alerts

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of existing literature and relevant research to establish the foundation for the “.....” project. It focuses on e-commerce growth in Kenya, price comparison challenges, web scraping techniques for real-time monitoring, and related tools/systems, while excluding unrelated areas such as general e-commerce logistics or non-scraping AI applications. The review critically evaluates prior works to identify methodologies, findings, limitations, and gaps that the current project aims to address.

The chapter is organized around the specific objectives outlined in Chapter One. It begins with an examination of consumer challenges in e-commerce price comparison (Objective i), followed by web scraping design and implementation (Objective ii), data processing/visualization techniques (Objective iii), and automation/notification mechanisms (Objective iv). Overall, the review demonstrates the project's relevance in bridging gaps in consumer-facing tools for the digital market.

2.2 Review of Objective One: Investigating Challenges and Limitations Faced by Kenyan Online Shoppers in Comparing Prices and Tracking Deals

Kenya's e-commerce sector has grown rapidly, with market value projected to reach US\$886 million in 2025 and platforms like Jumia and Kilimall dominating (Cysparks, 2025; Statista, 2025). Studies highlight price sensitivity as a key consumer behavior driver, with many shoppers comparing across sites to find value amid dynamic pricing and promotions (Murithi, 2024; ResearchGate, 2021 on Jumia Kenya).

However, manual comparison remains inefficient: consumers browse multiple tabs/apps, leading to decision fatigue and missed deals (Thomi, 2022). Dynamic pricing—frequent adjustments based on demand—exacerbates this, as users cannot easily track history or verify discounts (Impact Media, 2024; Pricefx, 2025). In Kenya, fragmented platforms (Jumia vs. Kilimall) and limited integrated tools force reliance on incomplete mental comparisons (Mamba Technologies, 2025).

Existing global tools like CamelCamelCamel and Keepa excel at Amazon price history/alerts but are marketplace-specific, lack Kenyan focus, and do not aggregate local sites (Keepa vs. CamelCamelCamel comparisons, 2025). Limitations include no cross-platform support for Jumia/Kilimall and minimal emphasis on regional consumer needs (e.g., M-Pesa integration or local promotions). Critically, these tools overlook Kenya-specific barriers like variable internet access and trust issues in online deals (USIU-Africa study, 2024).

This reveals a gap: no localized, automated tool for real-time cross-platform comparison in Kenya, motivating BeiSmart KE's focus on local platforms and user alerts.

2.3 Review of Objective Two: Designing and Developing a Python-Based Web Scraper for Real-Time Product Data Extraction

Web scraping enables price monitoring by extracting data from e-commerce sites (ResearchGate, 2023; Shaikh et al., 2023). Python libraries dominate: BeautifulSoup for static HTML parsing (efficient for simple structures like JD.com equivalents), Scrapy for large-scale crawling, and Selenium for dynamic/JavaScript content (essential for Jumia/Taobao-like sites) (Wepub, 2025; ResearchGate, 2025 on real-time systems).

Studies show hybrid approaches (Selenium + BeautifulSoup) improve efficiency under energy/resource constraints or anti-scraping measures (ResearchGate, 2026). Ethical challenges include site blocks (CAPTCHAs, rate limits), terms-of-service violations, and server strain (Infatica, 2025; Cheq.ai, 2024). Dynamic pricing sites complicate extraction due to JavaScript rendering and frequent changes (Akamai, 2025).

Existing price comparison projects (e.g., using Scrapy/BeautifulSoup) achieve real-time extraction but often ignore ethics (e.g., no robots.txt compliance) or Kenyan platforms (JETIR, 2024; IARJSET, 2025). Global tools like CamelCamelCamel rely on APIs or limited scraping, limiting adaptability to non-Amazon sites (Plott Data, 2025).

Critically, prior works lack emphasis on ethical, low-impact scraping for developing markets. “.....” addresses this by prioritizing rate limiting, user-agent rotation, and compliance while targeting Kenyan sites.

2.4 Review of Objective Three: Implementing Data Storage, Processing, and Visualization for Price History and Comparisons

Data from scraping requires cleaning, storage, and analysis. Pandas excels at processing (normalization, comparisons), while SQLite/MongoDB provides lightweight storage for timestamped records (Wepub, 2025). Visualization tools like Plotly enable interactive trends (e.g., price history charts), aiding decision-making (Shaikh et al., 2023).

Studies note challenges in handling inconsistent formats (currencies, promotions) and ensuring accuracy amid dynamic changes (JETIR, 2024). Global trackers like Keepa offer detailed history but are platform-locked and lack customizable dashboards for non-experts (Keepa, 2025 comparisons).

Limitations include scalability for real-time data and user-friendly presentation in resource-constrained environments. This gap—few localized, accessible visualization tools—supports “.....”s Pandas/Streamlit approach for Kenyan users.

2.5 Review of Objective Four: Integrating Automation and Notification Mechanisms for Timely Alerts

Automation via schedulers (e.g., APScheduler) enables periodic scraping and threshold-based alerts (email/SMS via smtplib/Twilio) (Wepub, 2025). Tools like CamelCamelCamel provide alerts but lack cross-platform scope and Kenyan integration (e.g., SMS for low-connectivity users) (42Signals, 2024).

Ethical dynamic pricing concerns (transparency, discrimination) highlight alert needs to counter exploitation (Washington University Law Review, 2022; Cloudseed, 2024). Prior systems often overlook timely, personalized notifications in emerging markets.

“.....” fills this by combining scheduling with alerts tailored to price-sensitive Kenyan consumers.

2.6 Concept Map / Conceptual Diagram

The conceptual framework derives from literature gaps: independent variables (e-commerce platform structures, dynamic pricing mechanisms, anti-scraping measures) influence dependent variables (price comparison efficiency, deal detection accuracy, user savings/time saved).

Key Concepts:

- Web Scraping Techniques (BeautifulSoup/Selenium) → Data Extraction Reliability.
- Ethical Compliance (rate limiting, robots.txt) → Sustainability.
- Data Processing (Pandas) & Visualization (Plotly/Streamlit) → Usability.
- Automation & Alerts (APScheduler) → Timeliness.

Relationships: Platform challenges drive scraping needs; ethical tools ensure reliable extraction; processed data enables visualization/comparisons; automation delivers alerts, closing the loop for consumer empowerment.

- Murithi, S. K. (2024). Influence of e-commerce adoption on consumer buying behavior of Jumia Kenya customers. USIU-Africa.
- Shaikh, A., et al. (2023). E-commerce Price Comparison Website Using Web Scraping. ResearchGate.
- Wepub. (2025). Research on Real-time E-commerce Price Comparison System Using Python Web Scraping Technology.